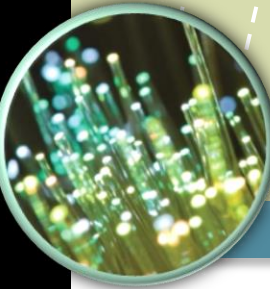


The Challenges of the Digital Age: Society & Information Technology Today

Chapter 9





Chapter Topics

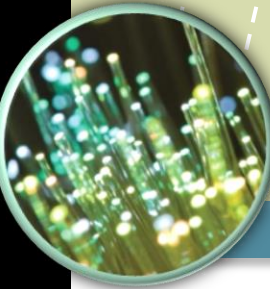
UNIT 9A: Security, Privacy, & Surveillance Concerns

- 9.1 Security Issues: Threats to Computers & Communications Systems
- 9.2 Security Safeguards: Protecting Computers & Communications
- 9.3 Privacy & Surveillance: Data Collection & Spies

UNIT 9B: Other Social, Economic, & Political Issues

- 9.4 Truth Issues: Manipulating Digital Data
- 9.5 Quality-of-Life Issues: The Environment, Mental Health, Child Protection, & the Workplace
- 9.6 Economic & Political Issues: Employment & the Haves/Have Nots





UNIT 9A: Security, Privacy & Surveillance Concerns

- Technology is now used to develop *predictive search apps* — applications that know what you want before you do. Some people, however, see the new technology as just the latest intrusion into our private lives, mining digital personal information about us whose uses we cannot always foresee.



9.1 Security Issues

Threats to Computers & Communication Systems



Errors, Accidents, & Natural Hazards

- **Human errors**
 - Humans often are not good at assessing their own information
 - Human emotions affect performance; people get frustrated
 - Human perceptions are slower than the equipment
 - Information overload may also be a problem

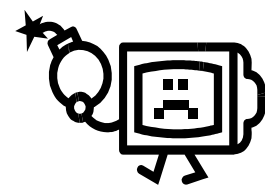




Procedural errors

- When people fail to follow established procedures, errors can occur

Software errors



- Software bug: an error in a program that causes it not to work properly

“Dirty data” problems

- Incomplete, outdated, or otherwise inaccurate data



GARBAGE IN,

GARBAGE OUT



Electromechanical problems

- Mechanical systems can wear out or become damaged
- They can also be badly designed or constructed
- Power failures and surges can damage equipment

Natural hazards can lead to disasters

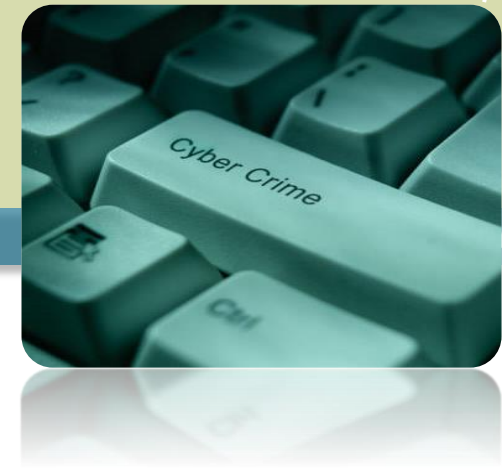


Computer Crimes



- Two types of computer crime:
 - Illegal act perpetrated against computers or telecommunications (computer is the target)
 - Use of computers or telecommunications to accomplish an illegal act (computer is the tool)





Computer Crimes (*continued*)



- Theft of hardware
- Theft of software
- Theft of intellectual property (piracy)
- Theft of time and services
- Theft of information (e.g., medical info, credit card info)
- Internet-related fraud (e.g., Wi-Fi phishing, or evil twin attack)
- Crimes of malice: crashing entire systems
 - Attacks on power-control systems and attacks on the Internet



9.2 Security Safeguards

Protecting Computers & Communications



Security is a system of safeguards for protecting information technology against disasters, system failures, and unauthorized access that can result in damage or loss.

- Computer security's five components:
 - Deterrents to computer crime
 - Identification and access
 - Encryption
 - Protection of software and data
 - Disaster recovery plans





Deterrents to computer crime

- Enforcing laws
- CERT: The Computer Emergency Response Team
 - Provides round-the-clock information on international computer security threats
- Tools to fight fraudulent and unauthorized online uses
 - Rule-based detection software
 - Predictive-statistical-model software
 - Employee Internet management software
 - Internet filtering software
 - Electronic surveillance

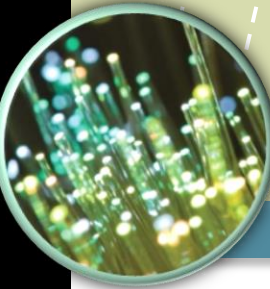




Identification and access

- Verify legitimate right of access by what you have, what you know, and who you are
 - What you have: cards, keys, signatures, badges
 - What you know: PINs and passwords; callback provision
 - Who you are: biometrics (such as hand geometry, fingerprint scans, iris recognition, face recognition, voice recognition)





Encryption



- The process of altering readable data into unreadable form to prevent unauthorized access
- Advantage: encrypting data that is available over the Internet keeps thieves and crackers from reading it
- Disadvantage: encrypting data may prevent law-enforcement officials from reading the data criminals are sending to one another

Discussion Questions: Does information privacy outweigh law enforcement's needs to track down and prosecute criminals? Should we all encrypt our information to prevent crackers and criminals from stealing it?



Protection of software and data



- Restrict access to online files; use transaction logs
- Use audit controls to track who used what programs and servers, which files were opened, and so on
- Use people controls—screen applicants; do background checks; monitor Internet, email, and computer usage; use shredders for discarded documents and materials



Disaster-recovery plans



- Method of restoring information-processing operations that have been halted by destruction or accident
- Back up everything; put mainframes in different locations
- Plans range in price and complexity
- Automatically store data redundantly in two or more places
- Have an off-site computer-based data storage center with independent power supply



9.3 Privacy & Surveillance

Data Collectors & Spies



The rise of Big Data has led to continuing threats to privacy from three giant sources:

- From business organizations
- From governments, local to federal
- From foreign governments and criminal groups





Business & Cyberspying

- Almost everything we do online is being scooped up and recorded for use by marketers, and it's difficult to know what parts of our own lives still belong to us.
- Whatever the impact on your personal privacy, it seems unlikely that you can claim ownership of a lot of data that's being collected about you. At work, for instance, you basically have no rights.



Government & Cyberspying

- Governments at all levels spy on their citizens, sometimes encouraged by the law, sometimes in spite of the law, often unknown to us.
 - Local police, national ID cards, National Security Agency (NSA), FBI, drones, and so on



Spying, Hacking, & Cyberwarfare by Foreign Governments & Groups

- The world is so interconnected that it is a constant struggle for technology managers to keep us secure against cyberinvasions of all sorts.
- Governments get involved in cyberwarfare —warfare involving computers and the Internet in which one nation attacks another's information systems.



UNIT 9B: Other Social, Economic, & Political Issues

- The changes brought about by technology are not all to the good. Now we consider some relevant social, economic, and political issues.

9.4 Truth Issues

Manipulating Digital Data



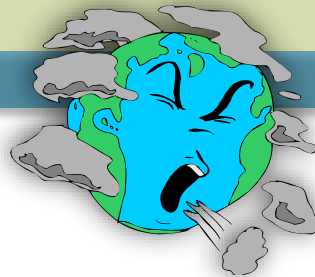
Digital sound and images can be manipulated (morphed).

- *Pros:*
 - Creates new forms of art
 - Digital technology allows musicians to sing every track of a song and accompany themselves
 - Morphing software makes one image morph into another
 - Movie special effects are made possible
 - Adobe Photoshop allows changes, enhancements to photos
- *Cons:*
 - Has made “recordings” (sound manipulation) and photographs (photo manipulation) untrustworthy
 - “News” can be faked



9.5 Quality-of-Life Issues

The Environment, Mental Health, Child Protection, & the Workplace



Environmental Problems

- Manufacturing computers and circuits can cause pollution
 - Hazardous toxins are involved in computer manufacture
 - Wireless devices can interfere in hospital activities and with medical devices
- Used computers/monitors contain chromium, cadmium, lead, mercury, PVC, and brominated flame retardants – all toxic substances that must be disposed of properly
- Visual pollution (“blight”) is created by the forest of wireless towers, roof antennas, satellite dishes, etc.; birds and bats, other wildlife, and vegetation are affected
- Nanotechnology carries possible risks on the molecular level



Mental-Health Problems

- *Isolation*: computer gamers may substitute online games for interpersonal interaction; videogame and Internet addiction
- Online gambling is too easy
- Many users find computers stressful and anger-inducing





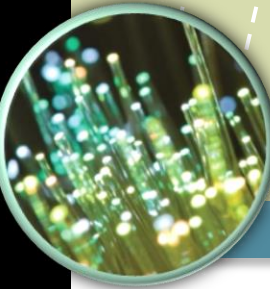
Protecting Children

- Pornography
 - The Internet has allowed the widespread distribution of sexually explicit material
 - Online pornographers use pop-up ads, Internet search engines to troll for new customers
 - This means that children may be exposed to porn when involved in innocent online searches
 - Online blocking software, DVD filters, the V-chip, and .xxx web addresses can be used to avoid/filter out pornography



Online Sexual Predators

- *Prevention Strategies*
 - Monitor children's Internet use
 - Be candid to children about threats
 - Caution children about revealing too much information
 - Tell them not to engage in **sexting** (online information and images NEVER go away; they are there forever)



Sexting is use of a smartphone or other mobile device to send sexually explicit photos or videos; sometimes it also refers to sexually charged text messages.

- Can lead to dangerous behavior and embarrassing experiences.





Cyberbullies

- Another example of how information technology can negatively affect children
- When it happens:
 - Save the evidence
 - Block messages
 - Contact an attorney or the police
- Cyberbullying can also occur in the workplace





Workplace Problems

- *Misuse of technology*
 - Playing online games, shopping, writing personal emails, etc. interfere with workers' productivity and can get them fired
- *Fussing with computers*
 - Dealing with hardware, software, and network connections can waste a lot of time, as can dealing with spam and the like
- *Information overload*
 - With so much available technology, people tend to work more hours and get swamped by too much information



Information Overload

- Term first used by Alvin Toffler in 1970: he predicted that the rapidly increasing amounts of information being produced would eventually cause people problems.
- Although computer processing and memory speed and capacity are increasing all the time, the brain that humans must use to process the information is not getting any faster.





What are some of the signs of information overload?

- Increased cardiovascular stress owing to a rise in blood pressure.
- Weakened vision.
- Confusion and frustration.
- Impaired judgment based on overconfidence.
- Irritation with others owing to an environmental input glut (which may also account for part of the “brusqueness” that is commonly attributed to big-city dwellers).





What can be done about information overload? For example:

- Spend less time on information that is nice to know and more time on information that you need to know now.
- Focus on getting relevant information, not on getting information faster, and focus on quality of information, rather than quantity.
- Learn how to create better information. Be direct in what you ask people, so that they can provide short precise answers.
- Understand the tools you have and don't switch tasks very often (single-tasking keeps the mind focused on one issue at a time).
- Avoid interruptions.
- Have quiet periods, when you disconnect.
- Take breaks.



9.6 Economic & Political Issues

Employment & the Haves/Have Nots



- Technology may replace humans in many jobs.
- Technology may affect the gap between the rich and the poor (“digital divide”).
 - Most jobs require employees who are tech-savvy.
 - People who are not tech-savvy won’t qualify for those jobs.
 - Technology is being used to replace employees in traditional jobs, traditionally filled by untrained workers.
 - Not everyone can afford computers and Internet connectivity.
- Many governments censor Internet content available in their countries, but they can’t control all Internet political activism.
- Internet is only loosely policed, so criminals take advantage.
- Being able to adapt to technology is critical for people.